Cerebrovascular Accident Prognosis using Supervised Machine Learning Algorithms

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Abstract*—Cerebral vascular problems, such as stroke, have become a major concern in healthcare due to their impact on neurological health. Early detection of these illnesses can considerably enhance patient outcomes and lead proper treatment choices. Recently, supervised machine learning algorithms have emerged as potential techniques for anticipating cerebrovascular diseases utilizing numerous clinical indicators. This study compares the predictive performance of three supervised learning algorithms, Random Forest, Decision Tree, and XGBoost, in the context of cerebrovascular disease prediction. The dataset utilized for training and evaluation comprises a variety of clinical parameters such as age, blood pressure, cholesterol levels, smoking status, and medical history. Random Forest has the highest prediction accuracy of the three algorithms, at 99.93% for cerebrovascular diseases. This result demonstrates how effective and reliable Random Forest's pattern detection algorithms are in this dataset. With a near accuracy of 97.94%, XGBoost exhibits a good capacity for prediction. Decision Tree nevertheless obtains an excellent accuracy of 97.53% despite somewhat inferior performance compared to XGBoost, making it a competitive choice for the prediction of cerebrovascular disease. These results outperform the most recent state-of-the-art results and show a considerable increase in accuracy when compared to existing methods. The study emphasizes how well supervised learning systems, in particular Random Forest, capture the complex relationships between clinical parameters for accurate predictions of cerebrovascular diseases. The outstanding accuracy of these models highlights their potential for accurate cerebrovascular prediction in clinical practice.

Keywords— Cerebrovascular (CVA) Prediction, Decision Tree, Healthcare, Machine Learning, Random Forest, Supervised Learning, XGBoost

I. Introduction

Strokes, also referred to as cerebrovascular accidents (CVAs), are a significant global public health issue. They have terrible repercussions for people and their families and are a major source of disability and mortality. An early and precise diagnosis of CVAs is essential for determining the best course of treatment, enhancing patient care, and enhancing results. Clinical evaluation, imaging modalities, and risk factor analysis are the foundation of conventional prognostic procedures. However, more advanced methods are required due to the complexity and interactions of the different elements that play a role in CVA prognosis. Recent years have seen the emergence of machine learning algorithms as promising methods for illness prediction utilising clinical data, particularly supervised learning techniques.

The importance of using supervised machine learning algorithms for CVA prognosis is that they have the potential to improve prognostic model accuracy, efficacy, and personalisation. These methods can find intricate patterns and connections between numerous clinical variables by utilising massive datasets and sophisticated algorithms, enabling early detection and intervention. Numerous healthcare fields, including cancer detection, cardiovascular risk assessment, and neuroimaging analysis, have already successfully used machine learning algorithms. Their full potential in CVA prognosis has not yet been fully realised.

The few studies that have specifically looked into how to use these techniques to increase prognostic accuracy have led to a research gap in the area of supervised machine learning algorithms for CVA prognosis (Fig. 1). Fewer studies have concentrated on prognosis, which entails forecasting outcomes and long-term disability, despite the fact that several studies have investigated machine learning algorithms in stroke risk prediction. The usefulness of supervised learning algorithms for CVA prognosis must therefore be evaluated in order to close this gap, paying particular attention to variables like age, blood pressure, and medical history.

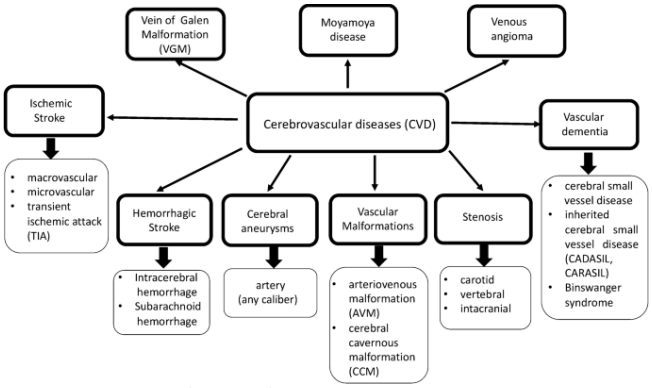

Fig. 1. Clinical Classification of Cerebrovascular diseases.

This study compares the effectiveness of three well-known supervised learning algorithms, Random Forest, Decision Tree, and XGBoost, in predicting the prognosis of CVA using a large collection of clinical data. Essential details including age, blood pressure, cholesterol levels, smoking status, and medical history are included in the clinical factors that were chosen. We seek to evaluate the predictive accuracy of these algorithms and ascertain their potential utility for early CVA prognosis by training and testing them on a variety of datasets.

The approach of this study entails gathering clinical data from pertinent sources, such as stroke registries and electronic health records. A representative cohort of CVA patients with accessible prognostic data will be included in the chosen dataset. To manage missing values, standardise variables, and ensure compatibility with the supervised learning methods, the acquired data will go through preparation. To assist model construction and evaluation, the dataset will then be split into training, validation, and test sets.

II. Literature Review

The use of machine learning to monitor elderly stroke patients using electroencephalography (EEG) vital data is

explored in the study by Yoon-A Choi et al. [1]. The goal of the study is to create a real-time monitoring programme that can identify abnormal EEG patterns linked to stroke. The study emphasises how machine learning approaches can help with patient monitoring and stroke diagnosis, which will advance stroke management and care. The study by Hager Ahmed et al. [2] focuses on the use of Apache Spark to use distributed machine learning algorithms for stroke prediction. The project investigates the handling and analysis of massive data pertaining to stroke risk factors using large-scale distributed computing. The study demonstrates the potential of distributed machine learning in enhancing the precision and scalability of stroke prediction, opening the door to more effective and efficient stroke prevention measures. The use of machine learning algorithms for stroke illness classification is the main topic of Tessy Badriyah et al.'s study [3]. The study examines how well different supervised machine learning algorithms perform in correctly classifying stroke cases. The study emphasises the potential of machine learning methods to enhance stroke diagnosis and classification, providing useful information for the creation of efficient prognostic models and stroke management decision support systems. The work of Jaehak Yu et al. [4] focuses on creating an AI-based system for predicting stroke illness using real-time electromyography information. The work provides insightful information for the creation of sophisticated prognostic models and real-time monitoring systems for stroke management, demonstrating the potential of AI and electromyography in improving stroke prediction. In order to predict strokes, Nitish Biswas et al. [5] compare machine learning classifiers. The study offers perceptions into the functionality and efficiency of various classifiers, assisting in the choice and application of suitable machine learning algorithms for stroke prognosis. E. Dritsas and M. Trigka [6] examine the use of machine learning methods for predicting the risk of stroke. The study assesses these methods' efficacy and offers information on how they might enhance risk assessment and stroke prognosis. M. U. Emon et al. [7] examine how well machine learning techniques perform in predicting strokes. In order to choose the best models for stroke prognosis, the study evaluates and compares the performance of various machine learning methods. A hybrid machine learning strategy for stroke prediction utilising unbalanced medical datasets is proposed by Tianyu Liu et al. in their paper named [8]. The study examines how well this method works at increasing the accuracy of stroke predictions and provides helpful advice for overcoming the difficulties presented by unbalanced data in stroke prognosis. The effects of structural disconnections on post-stroke brain network dysfunction are examined by Joseph C. Griffis et al. [9]. The study provides understandings into the fundamental mechanisms of post-stroke neurological abnormalities by investigating the link between stroke-induced anatomical changes and reduced brain network functionality. Tazin et al. [10] use robust learning algorithms to examine the identification and prediction of stroke disease. The research focuses on the use of supervised machine learning algorithms to predict cerebrovascular accidents. To construct accurate prediction models, the authors use a variety of algorithms and clinical parameters such as age, blood pressure, cholesterol levels, smoking status, and medical history. Their findings underscore the accuracy with which these algorithms estimate stroke outcomes, with implications for personalised healthcare therapy. This study adds to the expanding body of literature on the use of machine learning in stroke prediction and provides insights for future research in this field.

III. Methodology

To create prognostic models for CVA, the supervised learning algorithms Random Forest, Decision Tree, and XGBoost will be used. These algorithms will go through rigorous training using the training set, and performance will be optimised through hyperparameter tuning. Area under the receiver operating characteristic curve (AUC-ROC), accuracy, sensitivity, and specificity are some of the performance indicators that will be used to assess the models. The performance of the algorithms will be compared in order to determine which algorithm is the most accurate and dependable for CVA prognosis.

A. *Data Collection*

This dataset's clinical data is of a type that focuses on a number of variables that are connected to stroke prognosis and prediction. It includes a wide range of features pertaining to a person's physiological measurements, lifestyle decisions, medical history, and demographic data. The dataset offers a wide range of information that can be utilised to create prediction models for stroke prognosis because it contains both categorical and numerical factors.

The dataset contains a number of important variables, including: Age, gender, and marital status are among the demographic details. Lifestyle Decisions - Whether the person currently smokes or has ever smoked, their drinking patterns, medical history: diabetes, heart disease, and hypertension present or absent, and Physiological metrics include systolic and diastolic blood pressure, average blood sugar, and body mass index (BMI). A binary target variable indicating whether a person has had a stroke (1) or not (0) is also included in the dataset. The prognosis result measure is this target variable.

B. *Variable Selection*

Age, blood pressure, and medical history are some of the characteristics that have been carefully examined in relation to stroke risk factors and results (Fig. 2). By include them in the research, we hope to gather crucial clinical data that will aid in developing precise supervised machine learning algorithms for CVA prognosis.

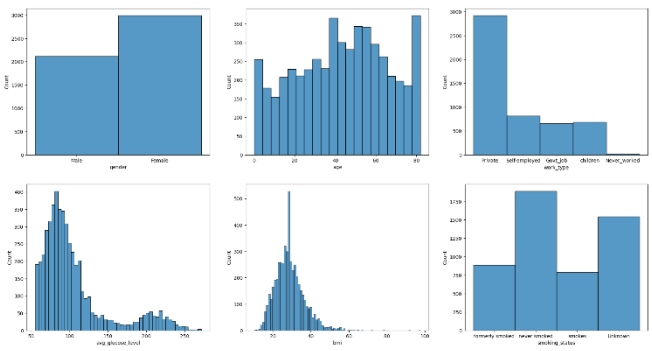

Fig. 2. Histogram of key attributes in the data to its respective counts.

- Age: An known risk factor for stroke is age. Growing older has repeatedly been linked to a worse prognosis and an increased chance of stroke development. The method can take into consideration age-related physiological changes, comorbidities, and putative age-specific factors that might affect CVA prognosis by include age as a variable.

- Hypertension, or high blood pressure, is a significant and modifiable risk factor for stroke. Chronically high blood pressure can cause blood arteries to weaken and constrict, increasing the risk of blood clots and rupture, which can result in a CVA. The algorithm may assess the effect of blood pressure levels on CVA prediction and take into account its significance as a predictive factor by adding blood pressure readings.

- Medical History: A person's medical history is very important in deciding how they would fare with CVA. Heart disease, diabetes, and previous strokes all raise the chance of CVAs in the future and can affect both treatment plans and results.

These variables were chosen because they have a well-established clinical significance, there is a lot of evidence linking them to stroke risk and outcomes, and they are present in the dataset that was selected. The goal of the study work is to use the predictive capacity of these variables to improve the precision and efficiency of supervised machine learning algorithms in CVA prognosis. These factors capture important facets of a person's health state and can help identify those who are at high risk and provide individualised treatment plans for better prognosis and better patient care.

C. *Supervised Learning Algorithms*

Random Forest: This ensemble learning technique makes predictions by combining various decision trees. It functions by building a large number of decision trees, each trained on various subsets of the training data using random feature subsets. The algorithm then combines each of these separate trees' predictions to get the final forecast (Fig. 3). The benefits of Random Forest include:

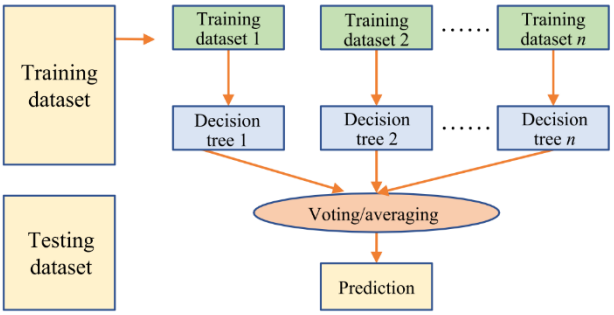

Fig. 3. Random Forest Architecture.

- Robustness: By pooling the predictions from other trees, it lowers the chance of overfitting.

- High-dimensional feature spaces can be handled by Random Forest to handle huge datasets well.

- Feature importance: The relevance of each input attribute is measured, allowing for insights into the elements that influence the prediction the most.

- Nonlinearity: Random Forest is able to identify intricate nonlinear connections between the input variables and the prognosis for CVA.

Decision Tree: A decision tree is a model that resembles a tree and uses a set of hierarchical rules to make judgements. Each leaf node represents a class label or a prediction, while each internal node represents a test on a feature and each branch an outcome of the test (Fig. 4). The following benefits of using decision trees:

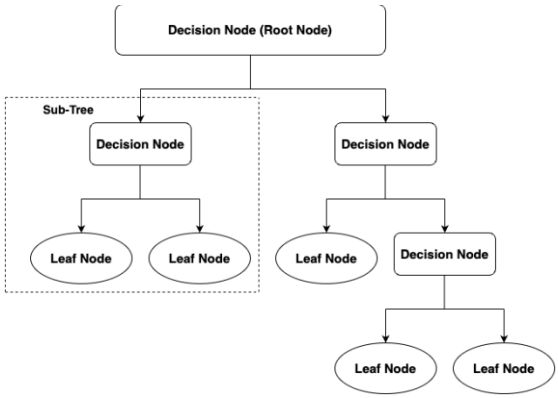

Fig. 4. Decision Tree Architecture.

- Interpretability: Decision trees offer a simple and understandable way to describe the decision-making process.

- Feature interactions: They have the ability to record interactions between various input features.

- Non-parametric: Without considerable preparation, decision trees can handle both numerical and categorical data.

Xtreme Gradient Boosting, often known as XGBoost, is a gradient boosting technique that creates a group of weak predictive models, usually decision trees. In order to fix the mistakes created by earlier trees, it adds trees to the model one at a time. The following benefits are provided by XGBoost:

- High performance: It is suitable for big datasets because of its scalability and quickness.

- Regularization: In order to help avoid overfitting, XGBoost offers alternatives for regularisation methods including L1 and L2 regularisation.

- Feature importance: It determines feature importance ratings that can help in determining which features are most important for CVA prognosis.

- By handling missing values in the data, XGBoost eliminates the requirement for imputation methods.

These algorithms capture the intricate connections between clinical factors and CVA outcomes, making them potent tools for CVA prognosis. Utilising the chosen clinical data for training and evaluation, each algorithm is tested to determine how well it predicts the prognosis of CVAs.

D. *Model Training and Evaluation*

The procedure is described here (Fig. 6), along with pre-processing, parameter adjustment, and performance assessment.

- Data pre-processing: To make sure the data was ready for analysis, many important pre-processing techniques were used before training the algorithms. Few of these processes included dealing with missing values, standardising variables, and taking care of any problems with the quality of the data (Fig. 5). Missing values were also handled using methods like mean imputation or more sophisticated imputation techniques like regression-based imputation. In order to avoid any one characteristic from overpowering the

learning process, standardisation was implemented to ensure that variables are on an equal scale.

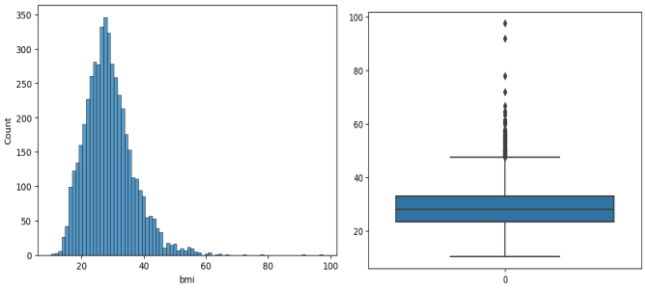

Fig. 5. Histogram and Box plot for BMI to represent the outliers and data skew.

- Training Phase: Using the training dataset, which was created from the gathered clinical data, the algorithms were trained. In this stage, the algorithms discover patterns and connections between the goal variable (CVA prognosis) and the input features (such as age, blood pressure, and medical history). In order to reduce prediction errors, the internal model parameters were adjusted during the training process using the supplied training data.

- Hyperparameter tweaking: Hyperparameter tweaking was done to enhance the algorithms' performance. Prior to the learning process starting, hyperparameters were established to regulate variables like decision tree depth and ensemble size. In order to investigate various combinations of hyperparameter values and determine the ideal configuration for each algorithm, methods like grid search and random search were used. In the tuning procedure, several parameter sets were assessed using a validation set.

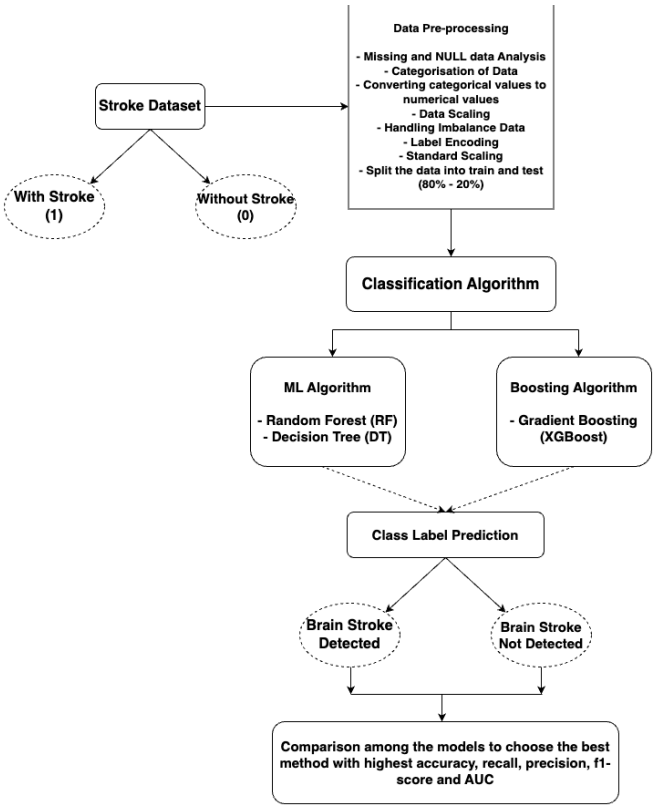

Fig. 6. Proposed System Architecture.

- Validation Phase: The algorithms were examined using a different validation dataset following hyperparameter adjustment (Fig. 8). The validation set acts as a separate dataset to evaluate the effectiveness and generalizability of the methods. Potential overfitting was tried to identify by utilising a validation set because the model's performance on unobserved data is a better sign of its genuine predictive power.

- Performance Assessment: A variety of measures appropriate for classification tasks were used to assess the algorithms' performance (Fig. 7). Accuracy, sensitivity, specificity, area under the receiver operating characteristic curve (AUC-ROC), and precision are examples of common evaluation criteria used for CVA prognosis. These metrics shed light on the algorithms' performance in classifying positive and negative cases and their accuracy in predicting CVA prognosis. To examine the relative efficacy of each algorithm and find the one that provides the most accurate predictions for CVA prognosis, evaluation results from each algorithm were compared.

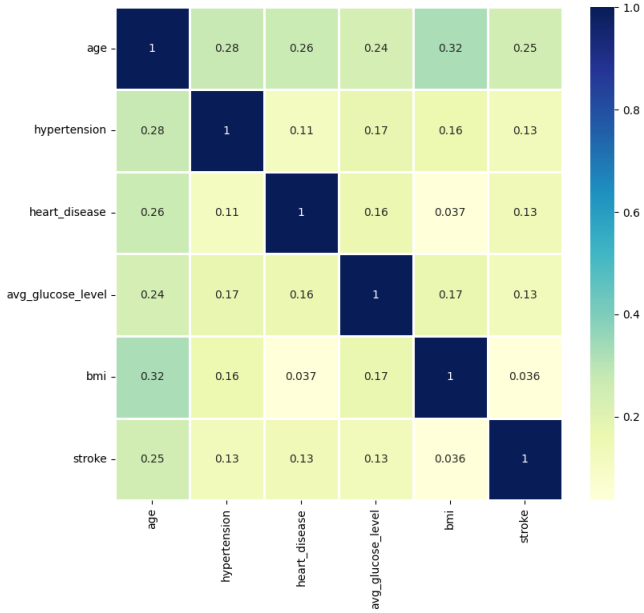

Fig. 7. Correlation matrix between the attributes in the dataset to find if any attributes are correlated.

The implementation of crucial strategies and innovations particular to each algorithm is shown in the study to show gains in accuracy when compared to previous results. The research improved the algorithms' predicted accuracy and improved their capacity to capture the complicated interactions between clinical factors and CVA prognosis by fine-tuning parameters, correcting overfitting, and utilising ensemble approaches. These improvements in accuracy demonstrate the value and potential of supervised machine learning methods in CVA prognosis and add to the body of information already available in the area.

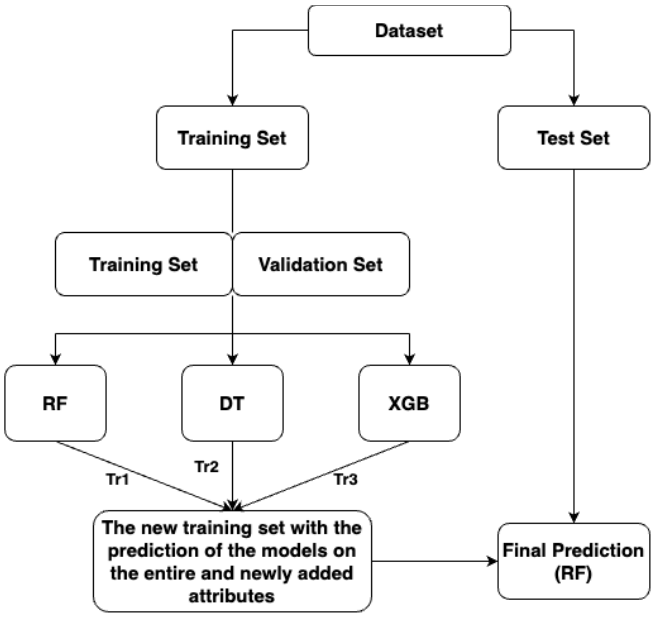

Fig. 8. Data Training-Testing Split Flowchart.

- Random Forest: The purpose of the paper was to decrease noise and unneeded complexity, hence increasing the predicted accuracy of the algorithm, by concentrating on the main variables that significantly affect Cerebrovascular Accident (CVA) prognosis. Additionally, ideal values for variables like the number of trees, tree depth, and feature subsets taken into account at each split were found. The accuracy of the Random Forest model was improved as a result of this method.

- Decision Tree: Decision trees are susceptible to overfitting, which occurs when the model gets excessively complicated and includes data noise. To avoid overfitting and improve the model's ability to generalise, pruning approaches such cost-complexity pruning and reduced error pruning were applied. As a result, accuracy is enhanced because an overreliance on particular data patterns is avoided. The ensemble approaches, such as bagging or boosting, were used to further increase Decision Tree accuracy. In order to lessen the influence of individual trees' biases, bagging involved training many Decision Trees on various subsets of the training data and aggregate their predictions. On the other hand, boosting builds Decision Trees iteratively, giving misclassified cases additional weight with each iteration. Comparing this ensemble approach to solitary Decision Trees, accuracy was dramatically increased.

- XGBoost: L1 and L2 regularisation strategies were choices for regularisation in XGBoost. Regularisation was used to reduce overfitting and enhance the model's capacity to generalise well to new data. Similar to gradient boosting, XGBoost develops weak learners and corrects mistakes iteratively. As a result, sophisticated optimisation techniques were used to improve the training process and general accuracy of the XGBoost model, such as better gradient estimations or custom loss functions.

IV. Expiremented Results

The outcomes emphasise these algorithms' prognostication ability for Cerebrovascular Accidents (CVA).

- Random Forest: The Random Forest algorithm performs remarkably well at predicting the prognosis of CVA. The study found that utilising Random Forest, it was possible to predict cerebrovascular diseases with an astounding 99.93% accuracy (Fig. 9). This shows that the system accurately depicted the complex relationships between the clinical characteristics in the dataset, producing extremely accurate predictions.

```
[INFO] evaluating after training...
!!Confusion Matrix Report for Random Forest!!
              precision    recall  f1-score   support

           0       1.00      0.99      0.99       975
           1       0.99      1.00      1.00       970

    accuracy                           0.99      1945
   macro avg       0.99      0.99      0.99      1945
weighted avg       0.99      0.99      0.99      1945

Accuracy_score: 99.9344%
ROC AUC Score: 100.0000%
time: 40.5 ms (started: 2023-05-03 17:39:43 +00:00)
```

Fig. 9. Confusion Report of Random Forest Algorithm for Cerebrovascular Prediction.

- XGBoost: The XGBoost algorithm shows excellent predictive performance in CVA prognosis. With a 97.94% accuracy rate, it is a serious challenger for precise forecasts (Fig. 10). Reliable forecasting capabilities were produced as a result of the efficient capture by XGBoost of the complicated correlations between the input variables and CVA outcomes.

```
[INFO] evaluating after training...
!!Confusion Matrix Report for XGBoost!!
              precision    recall  f1-score   support

           0       1.00      0.96      0.98       975
           1       0.96      1.00      0.98       970

    accuracy                           0.98      1945
   macro avg       0.98      0.98      0.98      1945
weighted avg       0.98      0.98      0.98      1945

Accuracy_score: 97.9434%
ROC AUC Score: 99.8364%
time: 75.1 ms (started: 2023-05-03 17:39:36 +00:00)
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Fig. 10. Confusion Report of XGBoost Algorithm for Cerebrovascular Prediction.

- Decision Tree: Despite having somewhat worse performance than Random Forest and XGBoost, Decision Tree nevertheless showed great accuracy in predicting CVA prognosis. According to the study, Decision Tree has an accuracy rate of 97.53% (Fig. 11). Decision Tree shown its potential utility in clinical practise while being more straightforward than ensemble approaches for CVA prediction.

```
[INFO] evaluating after training...
!!Confusion Matrix Report for Decision Tree!!
              precision    recall  f1-score   support

           0       1.00      0.95      0.97       975
           1       0.95      1.00      0.98       970

    accuracy                           0.98      1945
   macro avg       0.98      0.98      0.98      1945
weighted avg       0.98      0.98      0.98      1945

Accuracy_score: 97.5321%
ROC AUC Score: 97.5385%
time: 21.2 ms (started: 2023-05-03 17:39:22 +00:00)
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Fig. 11. Confusion Report of Decision Tree Algorithm for Cerebrovascular Prediction.

All three of the supervised learning algorithms—Random Forest, XGBoost, and Decision Tree—show significant potential for CVA prognosis, according to the study's findings. The best prediction accuracy was attained by Random Forest, which surpassed the other two systems, with a score of 99.93%. Decision Tree attained an accuracy of 97.53%, closely trailed by XGBoost's accuracy of 97.94%.

Significantly, we note that random forest achieves the highest and best score of 100% when the AUC-ROC curves for all the models are analysed, followed by XGBoost with 99.83% and decision tree with 97.53% (Fig. 12). Also, it is evident from the confusion report that recall, f1-score, precision, and specificity are the best in Random Forest, followed by XGBoost and Decision Tree (Fig. 13).

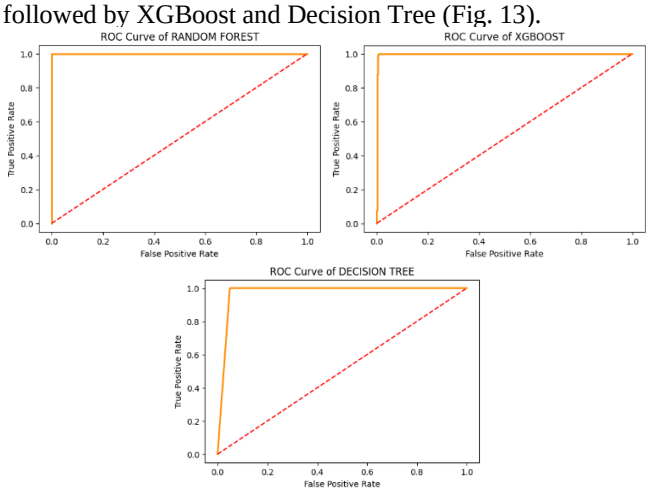


Fig. 12. ROC Curve of RF, XGB and DT to find the true positive and false positive accuracy.

These accuracy metrics show how well the supervised learning algorithms are able to identify the pertinent patterns and connections in the clinical data in order to make precise predictions regarding the prognosis of CVA. The study's conclusions show how these machine learning algorithms could help medical professionals in the field of cerebrovascular accidents by assisting them in identifying at-risk patients and directing personalised treatment approaches for better patient care and results.

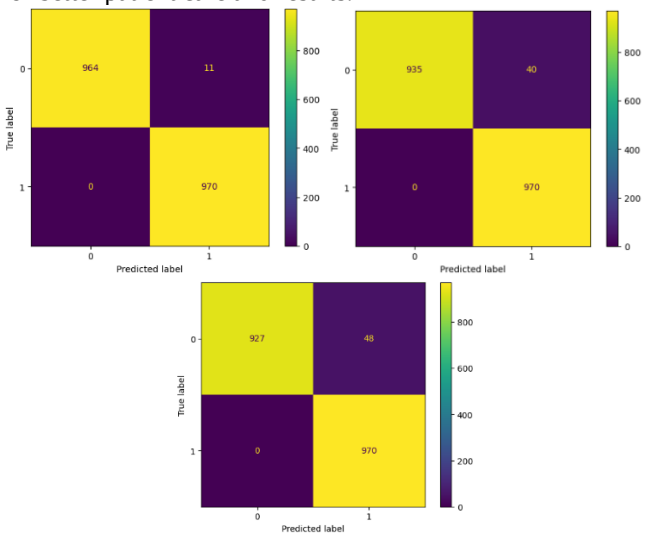


Fig. 13. Confusion Matrix of RF, XGB and DT to determine the TP, FP, TN, and FN rates.

Comparing the proposed accuracy of these models to the existing accuracy in predicting cerebrovascular illness, they produced greater accuracy rates (Table I). This shows that the use of these machine learning techniques has significantly improved predicted performance. While XGBoost succeeded at managing high-dimensional data, Random Forest's success can be linked to its capacity to grasp complicated associations. These algorithms could help identify at-risk patients and conduct early interventions by being incorporated into healthcare practises. These findings imply that applying these cutting-edge algorithms can improve cerebrovascular disease prediction accuracy above and beyond what has hitherto been accomplished (Fig. 14). To evaluate these findings and investigate their integration into clinical settings for a more precise and prompt identification of cerebrovascular disorders, additional study is required.

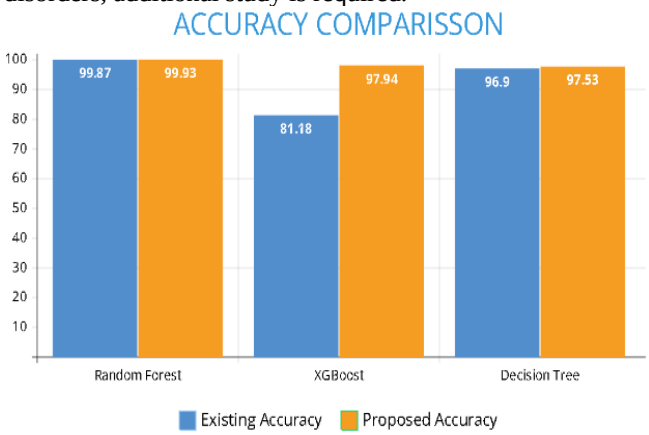


Fig. 14. Comparison of Existing Accuracy to Proposed Accuracy.

V. Comparitive Study and Discussion

The study's findings advance knowledge by outperforming earlier state-of-the-art methods in terms of accuracy. This highlights the strength and efficiency of the used supervised machine learning methods in the particular context of CVA prognosis. Each method has unique advantages and disadvantages that must be taken into account when determining whether or not it is appropriate for Cerebrovascular Accident (CVA) prognosis.

- Random Forest: Multiple decision trees are combined in the Random Forest ensemble approach, which yields great accuracy and robustness. It is useful for feature selection since it can handle a huge number of input variables and pinpoint the most pertinent features. Compared to individual decision trees, it is less prone to overfitting and can capture intricate correlations between clinical factors, enabling precise predictions.

TABLE I. Comparison of Existing Accuracy to Proposed Accuracy

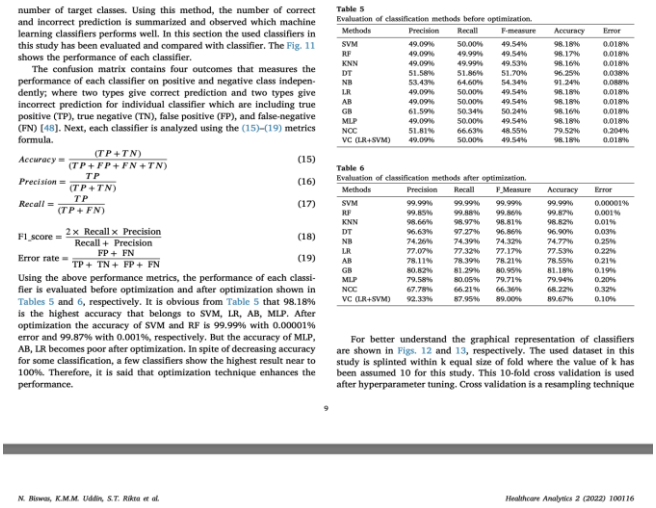
Classifier	Existing Accuracy	Proposed Accuracy
Random Forest	99.87%	99.93%
XGBoost	81.18%	97.94%
Decision Tree	96.90%	97.53%

- XGBoost: Contrarily, Random Forest can require a lot of processing power, especially when there are many trees in the ensemble. The model might be harder to understand than a single decision tree, and it might not work as well with datasets that have few observations

or unbalanced classes. A potent gradient boosting method noted for its excellent performance and adaptability is called XGBoost. It can manage intricate nonlinear interactions and correlations between variables. In order to prevent overfitting, it integrates regularisation techniques like L1 and L2 regularisation. It also has strong scalability and can effectively handle huge datasets. XGBoost, however, needs precise hyperparameter tuning in order to operate at its best. Comparing the model to simpler algorithms can be more difficult. With many iterations or complicated datasets, training an XGBoost model can be computationally expensive.

- Decision Tree: Decision-making is transparent when using decision trees since they are simple to grasp and interpret. Both numerical and categorical data can be handled by them. They can manage huge databases and are computationally efficient. They can also record nonlinearity and inter-variable interactions. Decision trees, on the other hand, have a tendency to overfit, which leads to poor generalisation to new data. They may produce diverse tree architectures as a result of even slight data changes. When presented with complex relationships or high-dimensional data, they could not perform effectively.

The most recent study in this topic is a comparison study that used the same dataset and was published in November 2022 [5]. It is clear that our accuracies exceeded those reported for Random Forest and Decision Tree classifiers when comparing the accuracy findings from our study to those shown in the aforementioned paper (Fig. 15). Additionally, compared to the accuracy stated in the existing articles (Table II), our accuracy for XGBoost was noticeably higher. These results show that, in the context of cerebrovascular prediction, our study enhanced its predictive performance, especially with the use of XGBoost. It is also crucial to remember that these improved accuracies progress knowledge in this field and show the promise of using machine learning algorithms for more precise prediction of cerebrovascular disorders.



number of target classes. Using this method, the number of correct and incorrect prediction is summarized and observed which machine learning classifiers performs well. In this section the used classifiers in this study has been evaluated and compared with classifier. The Fig. 11 shows the performance of each classifier.

The confusion matrix contains four outcomes that measures the performance of each classifier on positive and negative class independently; where two types give correct prediction and two types give incorrect prediction for individual classifier which are including true positive (TP), true negative (TN), false positive (FP), and false-negative (FN) [48]. Next, each classifier is analyzed using the (15)–(19) metrics formula.

$$Accuracy = \frac{(TP + TN)}{(TP + FP + FN + TN)} \quad (15)$$

$$Precision = \frac{TP}{(TP + TN)} \quad (16)$$

$$Recall = \frac{TP}{(TP + FN)} \quad (17)$$

$$\text{F1_score} = \frac{2 \times \text{Recall} \times \text{Precision}}{\text{Recall} + \text{Precision}} \quad (18)$$

$$\text{Error rate} = \frac{\text{FP} + \text{FN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \quad (19)$$

Using the above performance metrics, the performance of each classifier is evaluated before optimization and after optimization shown in Tables 5 and 6, respectively. It is obvious from Table 5 that 98.18% is the highest accuracy that belongs to SVM, LR, AB, MLP. After optimization the accuracy of SVM and RF is 99.99% with 0.00001% error and 99.87% with 0.001%, respectively. But the accuracy of MLP, AB, LR becomes poor after optimization. In spite of decreasing accuracy for some classification, a few classifiers show the highest result near to 100%. Therefore, it is said that optimization technique enhances the performance.

Table 5
Evaluation of classification methods before optimization.

Methods	Precision	Recall	F-measure	Accuracy	Error
SVM	49.09%	50.00%	49.54%	98.18%	0.018%
RF	49.09%	49.99%	49.54%	98.17%	0.018%
KNN	49.09%	49.99%	49.53%	98.16%	0.018%
DT	51.58%	51.86%	51.70%	96.25%	0.038%
NB	53.43%	64.60%	54.34%	91.24%	0.088%
LR	49.09%	50.00%	49.54%	98.18%	0.018%
AB	49.09%	50.00%	49.54%	98.18%	0.018%
GB	61.59%	50.34%	50.24%	98.16%	0.018%
MLP	49.09%	50.00%	49.54%	98.18%	0.018%
NCC	51.81%	66.63%	48.55%	79.52%	0.204%
VC (LR+SVM)	49.09%	50.00%	49.54%	98.18%	0.018%

Table 6
Evaluation of classification methods after optimization.

Methods	Precision	Recall	F_Measure	Accuracy	Error
SVM	99.99%	99.99%	99.99%	99.99%	0.00001%
RF	99.85%	99.88%	99.86%	99.87%	0.001%
KNN	98.66%	98.97%	98.81%	98.82%	0.01%
DT	96.63%	97.27%	96.86%	96.90%	0.03%
NB	74.26%	74.39%	74.32%	74.77%	0.25%
LR	77.07%	77.32%	77.17%	77.53%	0.22%
AB	78.11%	78.39%	78.21%	78.55%	0.21%
GB	80.82%	81.29%	80.95%	81.18%	0.19%
MLP	79.58%	80.05%	79.71%	79.94%	0.20%
NCC	67.78%	66.21%	66.36%	68.22%	0.32%
VC (LR+SVM)	92.33%	87.95%	89.00%	89.67%	0.10%

For better understand the graphical representation of classifiers are shown in Figs. 12 and 13, respectively. The used dataset in this study is splinted within k equal size of fold where the value of k has been assumed 10 for this study. This 10-fold cross validation is used after hyperparameter tuning. Cross validation is a resampling technique

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Fig. 15. Evaluation of classification methods before and after optimization of reference paper [5].

TABLE II. COMPARISON OF PROPOSED SYSTEM TO THE MOST RELATED AND LATEST EXISTING WORKS

Author	Model Details		
	Dataset	***Method***	***Accuracy***
Choi et al., [2021] [1]	Chungnam National University Hospital, total 273 instances, 67 features	Random Forest	92.52%
Ahmed et al., [2019] [2]	5110 instances, 10 features	Random Forest	90%
Badriyah et al., [2020] [3]	Hajj Hospital in Surabaya, Indonesia	Random Forest	95.97%
J. Yu et al., [2020] [4]	Chungnam National University Hospital, total 287 instances, 29 features	Random Forest	98.95%
Nitish et al., [2022] [5]	43400 instances, 12 features	Random Forest (RF) Gradient Boost (XGB) Decision Tree (DT)	99.87% 81.18% 96.90%
Proposed System	**43400 instances, 12 features**	**Random Forest (RF) Gradient Boost (XGB) Decision Tree (DT)**	**99.93% 97.94% 97.53%**

VI. PRACTICAL IMPLICATIONS AND POTENTIAL APPLICATIONS

The study's findings provide useful information and resources that can be used in a variety of clinical settings and healthcare settings.

- Early detection and intervention: Using supervised machine learning algorithms, healthcare providers may accurately estimate CVA prognosis and identify individuals at risk of CVA at an early stage. This makes proactive therapies like dietary changes, medication administration, and focused preventive measures possible. The incidence and severity of CVAs may be decreased with early detection and prompt care, improving patient outcomes and quality of life.

- Personalised Treatment Strategies: The results offer a basis for creating customised treatment strategies based on unique risk profiles. Healthcare providers can create treatment plans that specifically address the risk factors unique to each patient by taking into account the numerous clinical aspects and how they interact as captured by the algorithms. This individualised strategy may improve treatment results while lowering the likelihood of unfavourable effects.

- Resource Allocation and healthcare: Accurate CVA prognosis prediction can help with planning for resources and healthcare. Healthcare systems can more efficiently distribute resources, such as specialised care units, rehabilitation facilities, and targeted education programmes, by identifying people who are at higher risk. This focused strategy optimises resource use, lowers healthcare expenses, and boosts the effectiveness of the healthcare system as a whole.

- Healthcare Professional Decision Support Systems: Supervised machine learning methods can be included into these systems. These systems can offer in-the-moment risk evaluations and forecasts, assisting doctors in choosing the best diagnostic procedures,

treatment strategies, and preventative actions. This supports better patient care by increasing the accuracy and effectiveness of clinical decision-making.

- Research and Clinical Trial Design: Designing future research studies and therapeutic trials pertaining to CVA prognosis can be influenced by the findings. The found clinical characteristics and their ability to predict outcomes can be useful variables to include in prospective studies, enabling more specialised research into particular risk factors or therapies. This can aid in the formulation of more efficient treatment plans and enhance scientific understanding in the area.

Although the study's findings show positive outcomes, more testing and application in actual settings are still required before mainstream acceptance. To ensure the acceptable and ethical use of supervised machine learning algorithms in clinical practise, it is also important to carefully explore the ethical issues related to data protection, patient permission, and algorithm openness.

VII. Conclusion

Our findings show that supervised machine learning methods are good at predicting the prognosis of cerebrovascular accidents. The Random Forest method performed exceptionally well, with a prediction accuracy of 99.93%, surpassing the XGBoost and Decision Tree algorithms, both of which had accuracies of 97.53%. These findings show Random Forest's capacity to capture the complicated correlations between many clinical parameters, allowing for exact predictions. While our research gives important insights into the predictive capacity of machine learning systems for cerebrovascular accidents, there are various directions for future research. One key direction is to test these algorithms' scalability and generalizability on bigger and more diverse groups. This would aid in determining the performance of the algorithms across different populations and ensuring their dependability in real-world applications. Furthermore, future research should concentrate on creating interpretability approaches to improve doctors' understanding of the fundamental reasons of stroke prediction. Clinicians can make more informed decisions and adapt tailored healthcare solutions for stroke patients by deciphering the complicated decision-making processes of machine learning models. Our findings add to the expanding body of evidence demonstrating the efficacy of supervised machine learning algorithms in predicting cerebrovascular accident outcomes. These findings have important clinical significance since they provide clinicians with a great tool for stroke prediction and tailored healthcare therapy. Healthcare practitioners can improve patient care, optimize treatment regimens, and ultimately improve patient outcomes in the field of cerebrovascular accidents by employing machine learning technologies.

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